

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Opticrop : smart Agricultural production optimization engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports a limited number of farms and users	Performance decreases as users increase	High
2	Sensor data is stored on a single database	Storage becomes inefficient with large datasets	High
3	Basic security with limited authentication	Increased risk of unauthorized access	medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of farmers	Deploy cloud-based infrastructure with load balancing and auto-scaling
2	Data Storage	Stores sensor data in a local database	Migrate to cloud databases (Firebase, AWS, or Azure) with automatic backup
3	Performance	Processes recommendations on a single server	Use distributed processing and AI optimization for faster predictions
4	Security	Basic login authentication	Implement multi-factor authentication, encrypted communication, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Improve AI-based crop recommendation accuracy	3 Months	Better crop yield and decision-making
Phase 3	Integrate drone and satellite imagery for crop monitoring	6 Months	Early disease detection and precision farming
Phase 4	Develop Android and iOS mobile applications	9 Months	Easy access for farmers from anywhere